

S_4 Permutahedron Continuous-Journey Disassembly

Six-strand decomposition via colored journeys on the truncated octahedron

Précis prepared for S. J. Gates Jr.

Following written instructions dated 2026-08-26, the 24-vertex S_4 permutahedron has been decomposed into six four-vertex strands by tracing an 8-edge continuous colored journey from each vertex of the base hexagonal face. Each strand's four recorded vertices form a right coset of the Klein four-group $V_4 = \{e, (12)(34), (14)(23), (13)(24)\}$ inside S_4 . The six cosets partition all 24 vertices with no overlaps. An interactive 3D visualization implements the strand-by-strand separation with animated edge tracing. Three encoding decisions require confirmation before the artifact is marked publishable.

Algorithm

The procedure, as stated in the written instructions:

1. Start at a vertex on the base face and *record* it.
2. Travel along 2 links using **blue** and **red**, record the vertex reached.
3. Continue along 4 links in the sequence **green**, **blue**, **red**, **green**, record the vertex reached.
4. Continue along 2 links using **blue** and **red**, record the vertex reached.
5. The four recorded vertices are one of the six quartets.
6. Repeat from the next vertex on the base face until all six strands are extracted.

Encoding decisions

Color-generator map

Confirmed from the referenced screenshot:

Color	Generator	Transposition
Blue	s_1	(12)
Green	s_2	(23)
Red	s_3	(34)

Base face

The s_2, s_3 hexagonal face where position 1 is fixed at value 1, confirmed visually from the screenshot (base of the figure):

$$1234 \xrightarrow{s_2} 1324 \xrightarrow{s_3} 1342 \xrightarrow{s_2} 1432 \xrightarrow{s_3} 1423 \xrightarrow{s_2} 1243 \xrightarrow{s_3} 1234$$

Journey word

Three segments, eight edges total:

Segment	Generators	Length	Net permutation
J_1	$\text{BR} \quad (s_1, s_3)$	2	$(12)(34) = A$
J_2	$\text{GBRG} \quad (s_2, s_1, s_3, s_2)$	4	$(13)(24) = C$
J_3	$\text{BR} \quad (s_1, s_3)$	2	$(12)(34) = A$

Full word: $[s_1, s_3, s_2, s_1, s_3, s_2, s_1, s_3]$. Vertices are recorded at cumulative positions $\{0, 2, 6, 8\}$.

Klein four-group structure

The four recorded vertices of each strand form a right coset $\sigma \cdot V_4$ where:

Element	Permutation	Double transposition
e	1234	identity
A	2143	$(12)(34)$
AC	4321	$(14)(23)$
C	3412	$(13)(24)$

The six strands

#	Seed	Route	Quartet	Sector
1	1234	1234 $\xrightarrow{\text{B}}$ 2134 $\xrightarrow{\text{R}}$ 2143 $\xrightarrow{\text{G}}$ 2413 $\xrightarrow{\text{B}}$ 4213 $\xrightarrow{\text{R}}$ 4231 $\xrightarrow{\text{G}}$ 4321 $\xrightarrow{\text{B}}$ 3421 $\xrightarrow{\text{R}}$ 3412	P_6	VM3
2	1324	1324 $\xrightarrow{\text{B}}$ 3124 $\xrightarrow{\text{R}}$ 3142 $\xrightarrow{\text{G}}$ 3412 $\xrightarrow{\text{B}}$ 4312 $\xrightarrow{\text{R}}$ 4321 $\xrightarrow{\text{B}}$ 4231 $\xrightarrow{\text{R}}$ 2431 $\xrightarrow{\text{G}}$ 2413	P_3	VM
3	1342	1342 $\xrightarrow{\text{B}}$ 3142 $\xrightarrow{\text{R}}$ 3124 $\xrightarrow{\text{G}}$ 3214 $\xrightarrow{\text{B}}$ 2314 $\xrightarrow{\text{R}}$ 2341 $\xrightarrow{\text{G}}$ 2431 $\xrightarrow{\text{B}}$ 4231 $\xrightarrow{\text{R}}$ 4213	P_2	TM
4	1432	1432 $\xrightarrow{\text{B}}$ 4132 $\xrightarrow{\text{R}}$ 4123 $\xrightarrow{\text{G}}$ 4213 $\xrightarrow{\text{B}}$ 2413 $\xrightarrow{\text{R}}$ 2431 $\xrightarrow{\text{G}}$ 2341 $\xrightarrow{\text{B}}$ 3241 $\xrightarrow{\text{R}}$ 3214	P_4	VM1
5	1423	1423 $\xrightarrow{\text{B}}$ 4123 $\xrightarrow{\text{R}}$ 4132 $\xrightarrow{\text{G}}$ 4312 $\xrightarrow{\text{B}}$ 3412 $\xrightarrow{\text{R}}$ 3421 $\xrightarrow{\text{G}}$ 3241 $\xrightarrow{\text{B}}$ 2341 $\xrightarrow{\text{R}}$ 2314	P_1	CM
6	1243	1243 $\xrightarrow{\text{B}}$ 2143 $\xrightarrow{\text{R}}$ 2134 $\xrightarrow{\text{G}}$ 2314 $\xrightarrow{\text{B}}$ 3214 $\xrightarrow{\text{R}}$ 3241 $\xrightarrow{\text{G}}$ 3421 $\xrightarrow{\text{B}}$ 4321 $\xrightarrow{\text{R}}$ 4312	P_5	VM2

The six quartets contain all 24 vertices of S_4 exactly once. Each quartet is a right coset $\sigma \cdot V_4$, confirming the S_4/V_4 quotient structure.

Visualization

The implementation renders an interactive 3D permutahedron (truncated octahedron) with the following behavior:

- **Initial state:** all 24 vertices and 36 colored edges displayed in the orientation matching the referenced screenshot, with the base hexagonal face at the bottom.

- **Strand separation:** a single button click triggers each strand extraction. The 8 journey edges glow sequentially in their generator colors (blue, green, red), pausing at each of the four recorded vertices, then the strand's vertices separate outward from the polyhedron.
- **Reassembly:** returns all strands to the assembled state.
- **Overview:** extracts all six strands sequentially with a single click.

Questions requiring confirmation

Three encoding decisions are not uniquely determined by the written instructions and require explicit confirmation before the artifact is marked publishable.

Q1: Blue-Red ordering in outer segments

The instructions say “using blue and red links” for the 2-link outer segments (J_1 and J_3) without specifying order. We implemented Blue then Red (s_1 then s_3).

Because $s_1 = (12)$ and $s_3 = (34)$ act on disjoint positions, they commute: $s_1 s_3 = s_3 s_1 = (12)(34)$. The endpoint is identical either way, but the intermediate vertex differs:

Order	Path from 1234
B R (Blue, Red)	1234 $\rightarrow$ 2134 $\rightarrow$ 2143
R B (Red, Blue)	1234 $\rightarrow$ 1243 $\rightarrow$ 2143

Is Blue-then-Red the intended traversal order?

Q2: Base face traversal order

The instructions name 1234, then 1324, then “the next node on the base.” We continue around the hexagonal perimeter:

$$1234 \rightarrow 1324 \rightarrow 1342 \rightarrow 1432 \rightarrow 1423 \rightarrow 1243$$

Is this the intended sequence of seeds?

Q3: Overall correctness

Are the six quartets listed above correct?